MATHEMATICAL ANALYSIS 1
HOMEWORK 4

(1) Let $x_0 \in \mathbb{R}$. Recall that $I_r(x_0) = (x_0 - r, x_0 + r)$ is the neighborhood of x_0 of radius $r > 0$, that $I_r^*(x_0) = I_r(x_0) \setminus \{x_0\}$ is the deleted neighborhood of x_0 of radius r, and that $I_M(+\infty) = (M, +\infty)$ is the neighborhood of $+\infty$ of threshold $M > 0$.
   (a) Let $r_1, r_2 > 0$. Prove that $I_{r_1}(x_0) \cap I_{r_2}(x_0)$ is a neighborhood of x_0, and find its radius.
   (b) Let $r_1, r_2 > 0$. Prove that $I^*_{r_1}(x_0) \cap I^*_{r_2}(x_0)$ is a deleted neighborhood of x_0, and find its radius.
   (c) Let $M_1, M_2 > 0$. Prove that $I_{M_1}(+\infty) \cap I_{M_2}(+\infty)$ is a neighborhood of $+\infty$, and find its threshold.

(2) Consider the sequence $\{a_n\}_{n\in\mathbb{N}}$ defined by $a_n = \frac{n^2+(-1)^n n}{n^2+1}$.
   (a) Determine whether the sequence has a finite limit, has an infinite limit, or is indeterminate.
   (b) Prove your answer using the definition of the limit.

(3) Using the definition of the limit, prove that

$$\lim_{n\to\infty} \frac{n^2}{1-2n} = -\infty.$$

(4) Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Suppose that the limit $\lim_{n\to\infty} |a_n|$ exists. Does $\lim_{n\to\infty} a_n$ exist? If so, prove it. If not, give a counterexample.

(5) The proof of the Monotone Convergence Theorem (Theorem 5.1 in the lecture notes) is written only for the case of an *increasing* sequence. Prove the case of a *decreasing* sequence: let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence and suppose that there exists $N \in \mathbb{N}$ such that $a_{n+1} \le a_n$ for all $n > N$.
   (a) If $\inf\{a_n \mid n > N\} > -\infty$, then $\lim_{n\to\infty} a_n = \inf\{a_n \mid n > N\}$.
   (b) If $\inf\{a_n \mid n > N\} = -\infty$, then $\lim_{n\to\infty} a_n = -\infty$.

(6) Consider the sequence $a_n = \arctan\left(\frac{5n+6}{n+1}\right)$, $n \in \mathbb{N}$.
   (a) Is this a monotone sequence?
   (b) Find its infimum and supremum (explain your answer using the definitions of infimum and supremum).
   (c) Do the minimum and maximum exist? If so, what are they?
   (d) Does the limit $\lim_{n\to\infty} a_n$ exist? If so, what is it? *Hint: use Problem 5.*

(7) Let $\{a_n\}_{n\in\mathbb{N}}$ be a real-valued sequence. Suppose that it has two subsequences, $\{a_{n_k}\}_{k\in\mathbb{N}}$ and $\{a_{m_k}\}_{k\in\mathbb{N}}$, which converge to two *different* real numbers $\ell_1 \neq \ell_2$. Prove that $\{a_n\}_{n\in\mathbb{N}}$ has no limit (neither a finite limit nor $\pm\infty$). *Hint: argue by contradiction, and treat separately the cases* $a_n \to \ell \in \mathbb{R}$ *(take* $\varepsilon = \frac{|\ell_1-\ell_2|}{4}$*) and* $a_n \to \pm\infty$.

(8) Let $x_n = (-1)^n \frac{2n+1}{n+1}$, $n \in \mathbb{N}$.
   (a) Show that $-2 < x_n < 2$ for all $n \in \mathbb{N}$.
   (b) Using the definition of the limit, show that the subsequence of even terms $\{x_{2k}\}_{k\in\mathbb{N}}$ converges to 2, and that the subsequence of odd terms $\{x_{2k+1}\}_{k\in\mathbb{N}}$ converges to -2.
   (c) Use Problem 7 to conclude that the sequence $\{x_n\}_{n\in\mathbb{N}}$ is indeterminate.
   (d) All the terms x_{2k} belong to the open interval $(-2, 2)$, and the sequence $\{x_{2k}\}_{k\in\mathbb{N}}$ converges. Is its limit in $(-2, 2)$? Explain why this does not contradict Proposition 5.3 of the lecture notes, and what Proposition 5.3 says about the limit of $\{x_{2k}\}_{k\in\mathbb{N}}$ when we use the closed interval $[-2, 2]$ instead.

(9) Let $f(x) = \frac{2x^2-3x+1}{x^2+4}$. Find $\lim_{x\to+\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$, and prove your answers using the definition of the limit.

(10) Using the ε-δ definition, prove that $f(x) = 2x^2 + 3$ is continuous at $x_0 = 1$ (that is, $\lim_{x\to 1} f(x) = f(1) = 5$).

Homework 4 Solutions

(1) (a) Let $x \in \mathbb{R}$. Then $x \in I_{r_1}(x_0) \cap I_{r_2}(x_0)$ if and only if $|x - x_0| < r_1$ and $|x - x_0| < r_2$, which holds if and only if $|x - x_0| < \min\{r_1, r_2\}$. Hence

$$I_{r_1}(x_0) \cap I_{r_2}(x_0) = I_{\min\{r_1, r_2\}}(x_0),$$

which is a neighborhood of x_0 of radius $\min\{r_1, r_2\} > 0$ (the smaller of the two neighborhoods, since they are nested).

(b) Since $I_r^*(x_0) = I_r(x_0) \setminus \{x_0\}$, we have

$$I_{r_1}^*(x_0) \cap I_{r_2}^*(x_0) = \big(I_{r_1}(x_0) \setminus \{x_0\}\big) \cap \big(I_{r_2}(x_0) \setminus \{x_0\}\big) = \big(I_{r_1}(x_0) \cap I_{r_2}(x_0)\big) \setminus \{x_0\}.$$

By part (a) this equals $I_{\min\{r_1, r_2\}}(x_0) \setminus \{x_0\} = I^*_{\min\{r_1, r_2\}}(x_0)$, a deleted neighborhood of x_0 of radius $\min\{r_1, r_2\}$.

(c) A real number x belongs to $I_{M_1}(+\infty) \cap I_{M_2}(+\infty)$ if and only if $x > M_1$ and $x > M_2$, i.e. if and only if $x > \max\{M_1, M_2\}$. Hence

$$I_{M_1}(+\infty) \cap I_{M_2}(+\infty) = I_{\max\{M_1, M_2\}}(+\infty),$$

a neighborhood of $+\infty$ of threshold $\max\{M_1, M_2\} > 0$.

(2) (a) The sequence has a finite limit, namely $\lim_{n\to\infty} a_n = 1$.

(b) *Proof.* Let $\varepsilon > 0$. We look for $N_\varepsilon \in \mathbb{N}$ such that $|a_n - 1| < \varepsilon$ for all $n > N_\varepsilon$. We have

$$|a_n - 1| = \left|\frac{n^2 + (-1)^n n - (n^2+1)}{n^2+1}\right| = \frac{|(-1)^n n - 1|}{n^2+1} \le \frac{n+1}{n^2+1}.$$

For $n \ge 1$ we have $n + 1 \le 2n$ and $n^2 + 1 > n^2$, so

$$|a_n - 1| \le \frac{n+1}{n^2+1} < \frac{2n}{n^2} = \frac{2}{n}.$$

We want $\frac{2}{n} < \varepsilon$, i.e. $n > \frac{2}{\varepsilon}$. Choose $N_\varepsilon = \left\lceil \frac{2}{\varepsilon} \right\rceil$ (note that $N_\varepsilon \ge 1$ because $\varepsilon > 0$). Then for all $n > N_\varepsilon$ we have $n \ge 1$ and $n > \frac{2}{\varepsilon}$, so

$$|a_n - 1| < \frac{2}{n} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\lim_{n\to\infty} a_n = 1$. □

(3) *Proof.* Let $A > 0$. We look for $N_A \in \mathbb{N}$ such that $\frac{n^2}{1-2n} < -A$ for all $n > N_A$. Choose $N_A = \lceil 2A \rceil$ (note that $N_A \ge 1$ because $A > 0$). Let $n > N_A$. Then $n \ge 2$, so $2n - 1 > 0$ and, since $2n - 1 < 2n$,

$$\frac{n^2}{1-2n} = -\frac{n^2}{2n-1} < -\frac{n^2}{2n} = -\frac{n}{2}.$$

Moreover $n > N_A \ge 2A$ gives $-\frac{n}{2} < -A$. Hence $\frac{n^2}{1-2n} < -A$ for all $n > N_A$. Since $A > 0$ was arbitrary,

$$\lim_{n\to\infty} \frac{n^2}{1-2n} = -\infty.$$

□

(4) No, in general $\lim_{n\to\infty} a_n$ need not exist. Counterexample: $a_n = (-1)^n$. Then $|a_n| = 1$ for all n, so $\lim_{n\to\infty} |a_n| = 1$ exists. However $\{a_n\}_{n\in\mathbb{N}}$ does not converge: suppose by contradiction that $a_n \to \ell$. Taking $\varepsilon = 1$ in the definition, there is $N \in \mathbb{N}$ such that $|a_n - \ell| < 1$ for all $n > N$. Then for $n > N$,

$$2 = |a_n - a_{n+1}| \le |a_n - \ell| + |\ell - a_{n+1}| < 1 + 1 = 2,$$

a contradiction. The sequence is bounded, so it does not diverge either: it is indeterminate.

Remark. If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$ as well, because $\big||a_n| - 0\big| = |a_n - 0|$. So the answer is "yes" in the special case where the limit of $|a_n|$ is 0.

(5) *Proof.* Write $m = \inf\{a_n \mid n > N\}$. Note that for $n > N$ the sequence is decreasing, so $a_n \le a_k$ whenever $n \ge k > N$.

(a) Suppose $m > -\infty$ and fix $\varepsilon > 0$. Since $m + \varepsilon > m$ is not a lower bound of $\{a_n \mid n > N\}$, there exists $N_\varepsilon > N$ such that $a_{N_\varepsilon} < m + \varepsilon$. Let $n > N_\varepsilon$. Then $n > N$, so $a_n \ge m$ (as m is a lower bound), and $a_n \le a_{N_\varepsilon}$ (the sequence is decreasing for indices larger than N). Therefore

$$m \le a_n \le a_{N_\varepsilon} < m + \varepsilon,$$

and in particular $|m - a_n| < \varepsilon$ for all $n > N_\varepsilon$. By the definition of the limit, $\lim_{n\to\infty} a_n = m$.

(b) Suppose $m = -\infty$, i.e. $\{a_n \mid n > N\}$ is not bounded below, and fix $A > 0$. Then $-A$ is not a lower bound, so there exists $N_A > N$ such that $a_{N_A} < -A$. For every $n > N_A$ we have $a_n \le a_{N_A} < -A$. By the definition of divergence to $-\infty$, $\lim_{n\to\infty} a_n = -\infty$. □

(6) Write $b_n = \frac{5n+6}{n+1} = 5 + \frac{1}{n+1}$, so that $a_n = \arctan(b_n)$.

(a) If $n < k$ then $\frac{1}{n+1} > \frac{1}{k+1}$, so $b_n > b_k$; since arctan is strictly increasing, $a_n > a_k$. Hence the sequence is strictly decreasing, in particular monotone.

(b) *Supremum.* Since the sequence is decreasing, $a_n \le a_0 = \arctan 6$ for all n, so $\arctan 6$ is an upper bound that belongs to the set of values. Hence $\sup_n a_n = \arctan 6$.
Infimum. Since $b_n > 5$ for all n and arctan is strictly increasing, $a_n > \arctan 5$ for all n, so $\arctan 5$ is a lower bound. We show that it is the greatest lower bound. Let $L > \arctan 5$; we show that L is not a lower bound, i.e. that some $a_n < L$. If $L \ge \frac{\pi}{2}$ this is true for every n, since $a_n < \frac{\pi}{2}$. Otherwise $\arctan 5 < L < \frac{\pi}{2}$, and then $\tan L > 5$ (because tan is the inverse of arctan and is increasing on $(-\frac{\pi}{2}, \frac{\pi}{2})$). Choose $n \in \mathbb{N}$ with $n + 1 > \frac{1}{\tan L - 5}$. Then $b_n = 5 + \frac{1}{n+1} < \tan L$, so $a_n = \arctan(b_n) < \arctan(\tan L) = L$. Hence no number larger than $\arctan 5$ is a lower bound, and $\inf_n a_n = \arctan 5$. (The same argument works with $n \ge 1$, so also $\inf\{a_n \mid n > 0\} = \arctan 5$.)

(c) The maximum exists and equals $\sup_n a_n = \arctan 6$, attained at $n = 0$. The minimum does not exist: the infimum $\arctan 5$ is not attained, since $a_n > \arctan 5$ for all n.

(d) Yes. The sequence is decreasing for all $n > 0$ and $\inf\{a_n \mid n > 0\} = \arctan 5 > -\infty$, so by Problem 5 (with $N = 0$) the limit exists and

$$\lim_{n\to\infty} a_n = \inf\{a_n \mid n > 0\} = \arctan 5.$$

(7) *Proof.* Let $\ell_1 \ne \ell_2$ be the limits of the subsequences $\{a_{n_k}\}_{k\in\mathbb{N}}$ and $\{a_{m_k}\}_{k\in\mathbb{N}}$, where $\{n_k\}_{k\in\mathbb{N}}$ and $\{m_k\}_{k\in\mathbb{N}}$ are strictly increasing sequences of natural numbers (the indices of a subsequence always satisfy $n_{k+1} > n_k$, since each term of $\{a_n\}_{n\in\mathbb{N}}$ is used at most once and in order). In particular $n_k \ge k$ for all $k \in \mathbb{N}$, by induction: $n_0 \ge 0$, and if $n_k \ge k$ then $n_{k+1} \ge n_k + 1 \ge k + 1$. In the same way $m_k \ge k$ for all k. Suppose by contradiction that $\{a_n\}_{n\in\mathbb{N}}$ has a limit.

Case 1: $a_n \to \ell \in \mathbb{R}$. Let $\varepsilon = \frac{|\ell_1 - \ell_2|}{4} > 0$. By the definition of the limit there is $N \in \mathbb{N}$ such that $|a_n - \ell| < \varepsilon$ for all $n > N$. Since $a_{n_k} \to \ell_1$ and $a_{m_k} \to \ell_2$, there is $K \in \mathbb{N}$ such that $|a_{n_k} - \ell_1| < \varepsilon$ and $|a_{m_k} - \ell_2| < \varepsilon$ for all $k > K$. Fix $k > \max\{N, K\}$. Then $n_k \ge k > N$ and $m_k \ge k > N$, so $|a_{n_k} - \ell| < \varepsilon$ and $|a_{m_k} - \ell| < \varepsilon$. By the triangle inequality,

$$|\ell_1 - \ell_2| \le |\ell_1 - a_{n_k}| + |a_{n_k} - \ell| + |\ell - a_{m_k}| + |a_{m_k} - \ell_2| < 4\varepsilon = |\ell_1 - \ell_2|,$$

a contradiction.

Case 2: $a_n \to +\infty$. Let $A = \max\{1, \ell_1 + 1\}$. There is $N \in \mathbb{N}$ such that $a_n > A$ for all $n > N$. Since $a_{n_k} \to \ell_1$, there is $K \in \mathbb{N}$ such that $|a_{n_k} - \ell_1| < 1$, and so $a_{n_k} < \ell_1 + 1 \le A$, for all $k > K$. For $k > \max\{N, K\}$ we have $n_k \ge k > N$, hence $a_{n_k} > A$ as well, a contradiction.

Case 3: $a_n \to -\infty$. Let $A = \max\{1, 1 - \ell_1\}$. There is $N \in \mathbb{N}$ such that $a_n < -A$ for all $n > N$. Since $a_{n_k} \to \ell_1$, there is $K \in \mathbb{N}$ such that $a_{n_k} > \ell_1 - 1 \ge -A$ for all $k > K$. For $k > \max\{N, K\}$ we have $n_k > N$, hence $a_{n_k} < -A$ as well, a contradiction.

In all cases we reached a contradiction, so $\{a_n\}_{n\in\mathbb{N}}$ has no limit. □

(8) Note that $\frac{2n+1}{n+1} = 2 - \frac{1}{n+1}$.

(a) For all $n \in \mathbb{N}$, $1 \le 2 - \frac{1}{n+1} < 2$, hence $1 \le |x_n| < 2$ and so $-2 < x_n < 2$.

(b) For the even terms, $x_{2k} = 2 - \frac{1}{2k+1}$. Let $\varepsilon > 0$ and choose $K = \left\lceil \frac{1}{2\varepsilon} \right\rceil$. For $k > K$ we have $2k+1 > 2K+1 > \frac{1}{\varepsilon}$, so

$$|2 - x_{2k}| = \frac{1}{2k+1} < \varepsilon.$$

Hence $x_{2k} \to 2$. For the odd terms, $x_{2k+1} = -2 + \frac{1}{2k+2}$. With the same K and $k > K$ we have $2k+2 > \frac{1}{\varepsilon}$, so

$$|-2 - x_{2k+1}| = \frac{1}{2k+2} < \varepsilon,$$

and hence $x_{2k+1} \to -2$.

(c) By part (b), $\{x_n\}_{n\in\mathbb{N}}$ has the subsequences $\{x_{2k}\}_{k\in\mathbb{N}}$ and $\{x_{2k+1}\}_{k\in\mathbb{N}}$ converging to the two different numbers 2 and -2. By Problem 7, $\{x_n\}_{n\in\mathbb{N}}$ has no limit: it neither converges nor diverges to $\pm\infty$, i.e. it is indeterminate. (By part (a) the sequence is bounded, so the Bolzano–Weierstrass theorem guarantees that it has convergent subsequences; here we found two of them with different limits.)

(d) The limit of $\{x_{2k}\}_{k\in\mathbb{N}}$ is 2, which is *not* in $(-2, 2)$. This does not contradict Proposition 5.3, because that proposition concerns sequences in a *closed* interval $[a, b]$; the open interval $(-2, 2)$ is not of that type, and the statement is false for it. For the closed interval $[-2, 2]$, all the terms x_{2k} belong to $[-2, 2]$, and Proposition 5.3 guarantees that the limit belongs to $[-2, 2]$: indeed $2 \in [-2, 2]$. (The same holds for $\{x_{2k+1}\}_{k\in\mathbb{N}}$, with limit $-2 \in [-2, 2]$.)

(9) *Proof.* Note first that $f(x) - 2 = \frac{2x^2-3x+1-2(x^2+4)}{x^2+4} = \frac{-3x-7}{x^2+4}$. We claim that

$$|f(x) - 2| < \frac{10}{|x|} \qquad \text{for all } x \in \mathbb{R} \text{ with } |x| > 1.$$

Indeed, if $|x| > 1$ then $|-3x-7| \le 3|x| + 7 < 3|x| + 7|x| = 10|x|$ and $x^2 + 4 > x^2 = |x|^2$, so

$$|f(x) - 2| = \frac{|-3x-7|}{x^2+4} < \frac{10|x|}{|x|^2} = \frac{10}{|x|}.$$

Now fix $\varepsilon > 0$.

$x \to +\infty$**:** Let $N_\varepsilon = \max\left\{1, \frac{10}{\varepsilon}\right\}$. For all $x > N_\varepsilon$ we have $|x| > 1$ and $x > \frac{10}{\varepsilon}$, so $|f(x) - 2| < \frac{10}{x} < \varepsilon$. Hence $\lim_{x\to+\infty} f(x) = 2$.

$x \to -\infty$**:** Let $N_\varepsilon = -\max\left\{1, \frac{10}{\varepsilon}\right\}$. For all $x < N_\varepsilon$ we have $|x| > 1$ and $|x| > \frac{10}{\varepsilon}$, so $|f(x) - 2| < \frac{10}{|x|} < \varepsilon$. Hence $\lim_{x\to-\infty} f(x) = 2$. □

(10) *Proof.* Let $\varepsilon > 0$. Note that $f(1) = 5$ and

$$|f(x) - f(1)| = |2x^2 + 3 - 5| = 2|x^2 - 1| = 2|x-1|\,|x+1|.$$

If $|x-1| < 1$ then $0 < x < 2$, so $1 < x+1 < 3$ and $|x+1| < 3$. Choose $\delta = \min\left\{1, \frac{\varepsilon}{6}\right\}$. Then for $|x-1| < \delta$ we have both $|x-1| < \frac{\varepsilon}{6}$ and $|x+1| < 3$, so

$$|f(x) - f(1)| = 2|x-1|\,|x+1| < 2 \cdot \frac{\varepsilon}{6} \cdot 3 = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, f is continuous at $x_0 = 1$, i.e. $\lim_{x\to1}(2x^2 + 3) = 5$. □